MIL-T-21092C(YD)
13 April 1981
SUPERSEDING
MIL-T-21092B(YD)
27 January 1972

MILITARY SPECIFICATION

TRAP, AIR AND WATER, BALL FLOAT (150 PSIG WORKING PRESSURE)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single type of cylindrical, elliptical, or ball float trap. The trap shall operate automatically to vent air and other noncondensable gases or to drain water or oil from fluid systems.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:46Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B1.1 - Unified Screw Threads (UN and UNR Thread Form).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 Description. The float trap, when used as a vent, shall automatically vent air and other noncondensable gases from fluid systems or equipment. When utilized as a drain trap, the float trap shall automatically drain liquid from air and gas systems or equipment.

3.2 First article. When specified (see 6.2), the contractor shall furnish a float trap for first article inspection and approval (see 4.3.1 and 6.3).

* 3.3 Standard commercial product. Float traps of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the float trap being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 20260-19T08:46Z
Check the source to verify that this is the current version before use.

* 3.5 Interchangeability. All traps of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

* 3.6 Design. The equipment and accessories shall be so designed as to permit automatic operation that is satisfactory for its intended purpose. A safety factor of at least twice the operating pressure shall be incorporated in the equipment design to provide adequate protection to the equipment system in the event of failure. Traps of the open upright float type shall drain liquids from air or gas systems. Traps shall be float and valve mechanisms of positive snap-action, and shall be designed that will permit opening of the valve in the event of float failure.

3.7 Construction.

3.7.1 Body and cap. The body and cap shall be cast iron, cast steel, cast bronze, or forged steel suitable for the intended purpose.

3.7.2 Valve and valve seat. The valve and valve seat shall be corrosion-resistant material not affected by liquids or gases encountered. The valve seats may be made of acrylonitrile-butadiene elastomer, teflon, or its equivalents.

3.7.3 Float. The float may be round, cylindrical, or elliptical, shall be corrosion-resistant material, and shall be capable of sustaining, without damage, the pressures specified in 3.8.2.2.

3.7.4 Gasket. The gasket shall be capable of withstanding, without damage, the environmental conditions specified in 3.8.1. The material used shall not provide a nutrient for fungus and insects.

3.7.5 Inlet and outlet ports. The inlet and outlet ports shall be threaded with 3/4-14NPT female thread.

3.7.5.1 Port location. When venting air, the inlet port should be at the bottom of the trap; when water is drained, the inlet should be located on the side or at the top of the trap.

3.7.6 Valve. The valve shall be positioned or located in the body with sufficient clearance and in a manner that permits proper valve operation in the event of excess accumulation of sediment, scale, or other foreign material settling in the bottom of the trap.

3.7.7 Back vent port. Unless otherwise specified (see 6.2), the body or cap shall be provided with a boss drilled and tapped with 18NPT female thread of not less than 1/4 inch or more than 3/4 inch pipe sizes. The vent port shall be provided with a plug in the event back-venting of equipment is required.

3.7.8 Threads. Screw threads for all thread securing or attaching devices and threaded parts shall be in conformance with ANSI B1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:46Z
Check the source to verify that this is the current version before use.

3.7.9 Miscellaneous components. All internal parts, such as nuts, bolts, linkages, pins, etc., shall be made of corrosion-resistant materials suitable for the intended purpose specified in 6.1. All external components, not corrosion-resistant, shall be treated to resist corrosion.

3.8 Performance.

3.8.1 Humidity. The equipment, with ports blanked-off, when subjected to relative humidity ranging up to 95 percent and maintained at 90 percent ±5 percent, in a sea-air atmosphere during 72 hours, shall show no evidence of corrosion fungus, or other deterioration indicative of improper or faulty material or treatment.

3.8.2 Pressure.

3.8.2.1 Working pressure. The equipment shall operate satisfactorily throughout the differential pressure range from 5 pounds per square inch gage (psig) to 150 psig with the outlet vented to atmospheric pressure at 70° Fahrenheit (F).

3.8.2.2 Maximum pressure. The equipment shall be capable of sustaining an internal hydrostatic pressure of 300 psig ±10 psig for 15 minutes without leaks, permanent set, rupture, or other evidence of failure.

3.8.3 Capacity. The capacity of the trap shall be determined with equipment at 70 ±3.6°F and relative humidity of 50 ±5 percent. The trap outlet shall vent to the atmosphere. The minimum capacities specified in 3.8.3.1 and 3.8.3.2 shall be met with a single orifice (valve).

3.8.3.1 Water discharge. Unless otherwise specified (see 6.2), the minimum water-discharge capacity shall be 690 pounds per hour based on tap water subjected to a differential pressure of 147 ±2 psig at the orifice.

3.8.3.2 Air venting. Unless otherwise specified (see 6.2), the minimum air-venting capacity shall be 20 cubic feet of air per minute when tested as specified in 4.6.1.3.

3.9 Treatment and painting. Unless otherwise specified (see 6.2), all external surfaces of the body and cap exposed to the atmosphere shall be treated and painted in accordance with the manufacturer's best standard practice.

3.10 Identification marking. Equipment, assemblies, and parts shall be marked for identification in accordance with the manufacturer's standard practice.

* 3.11 Workmanship.

3.11.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:46Z
Check the source to verify that this is the current version before use.

3.11.2 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

3.11.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. All traps offered for delivery at one time shall be considered a lot for purposes of inspection.

4.3 Classification of inspections. Inspections shall be classified as follows:

(a) First article inspection (see 4.3.1).
(b) Quality conformance inspection (see 4.3.2).
* (c) Packaging inspection (see 4.7).

4.3.1 First article inspection. First article inspection shall be performed on one trap when required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and 4.7 and the tests of 4.6. The first article may be a first production item or a standard production item from the contractor's current inventory provided the trap meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining trap to be furnished under the contract.

* 4.3.2 Quality conformance inspection. The quality conformance inspection shall include the requirements of 4.5, 4.6, and 4.7.

4.4 Sampling.

4.4.1 Sampling for examination. A random sample of traps shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. The Acceptable Quality Level (AQL) shall be 1.5 percent defective units for major defects and 6.5 percent defective units for minor defects.

4.4.2 Sampling for tests. A random sample of traps shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be based on 6.5 percent defective units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:46Z
Check the source to verify that this is the current version before use.

* 4.5 Examination. Each trap selected in accordance with 4.4.1 shall be examined for compliance with the requirements specified in Table I. Any redesign of modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. Any trap in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample the lot represented by the sample shall be rejected.

TABLE I. Classification of defects.

Categories	Defects	Reference paragraph
Major:		
101	Parts not new.	3.4
102	Identical items not interchangeable.	3.5
103	Not of proper design.	3.6
104	Sticking or binding of moving parts.	3.6
105	Material not as specified.	3.7.1 through 3.7.3
106	Threaded parts not in accordance with ANSI B1.1.	3.7.8
107	Components missing.	3.7.9
Minor:		
201	Treatment and painting not as specified.	3.9
202	Parts not marked as specified.	3.10
203	Surfaces burred, nicked, or containing sharp edges.	3.11

4.6 Tests. When required, the first article (see 3.2), and each sample selected in accordance with 4.4.2 shall be tested to verify compliance with this specification. Tests shall be conducted in accordance with 4.6.1 through 4.6.1.4. Any unit failing to pass these tests shall be rejected and if the number of rejected units in any sample is equal to or greater than the rejected number for that sample, the lot represented by the sample shall be rejected.

4.6.1 Test conditions. Prior to conducting the outlined tests, each sample trap shall be subjected to a temperature between -80°F and -70°F for 72 hours; between +150°F and +160°F for 72 hours; and a relative humidity condition between 85 percent and 95 percent between +65°F and +75°F in a sea-air atmosphere for 72 hours. The trap shall then be examined for evidence of damage and shall meet the following tests:

4.6.1.1 Hydrostatic pressure test. The hydrostatic pressure test shall be an internal-pressure test with cold water. Traps shall be tested at a pressure of 300 psig ±10 psig for 15 minutes.

4.6.1.2 Cold water operational and capacity test. The cold water operational test shall consist of at least three runs spaced to cover the entire pressure range (see 3.8.2.1) to determine that proper drainage occurs. The capacity test shall be conducted at a differential pressure between 145 psig

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:46Z
Check the source to verify that this is the current version before use.

and 150 psig to determine that the water discharge capacity meets the requirements outlined in 3.8.3.1. For this test, the trap shall be connected to a pressure line or vessel and discharged through a water flow meter or into a weighing tank.

4.6.1.3 Air-venting operational and capacity test. The air-venting operational test shall consist of at least three runs spaced to cover the entire pressure range between 5 psig and 150 psig to determine that proper venting occurs. The capacity test shall be conducted at a differential pressure between 145 psig and 150 psig to determine that the air-discharge capacity meets the requirements outlined in 3.8.3.2. For this test, the trap shall be connected to an air supply and the discharge shall be measured by a suitable air flow meter or calculated either according to the change in pressure of the air supply or by the volume displacment method.

4.6.1.4 Endurance test. A limited endurance test of 8 hours or more, to cover the entire pressure range of the trap, shall be run on each sample to determine its ability to operate continuously for that period without damage or attention. During this test, the tightness of valve mechanism and its ability to handle the required amount of liquid or gas throughout the pressure range regularly and automatically shall be observed.

4.7 Packaging inspection. The preservation, packaging, packing, and marking of the traps shall be inspected to determine conformance with the applicable requirements of Section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. All unpainted exterior ferrous metal surfaces shall be coated with type P-1 preservative in accordance with MIL-P-116. Inlets and outlets shall be sealed with plastic plugs or threaded metal plugs.

5.1.2 Commercial. The equipment shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. The traps shall be packed in cleated plywood box, conforming to PPP-B-601 (overseas type). Contents of the boxes shall be cushioned, anchored, braced, and blocked in accordance with MIL-STD-1186. Flat steel strapping shall be type I, class B. The gross weight of wood-cleated boxes shall not exceed 200 pounds, except when the weight of the individual unit exceeds this amount. Wood boxes exceeding 200 pounds gross weight shall be modified by the addition of a minimum of two nominal 2- by 4-inch skids placed flat at right angles to the long dimension. Ends of the skids shall be beveled full depth at a 45° angle.

5.2.2 Level B. The traps shall be packed in a cleated plywood box conforming to PPP-B-601 (domestic type) or fiberboard box conforming to

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:46Z
Check the source to verify that this is the current version before use.

PPP-B-636, weather-resistant, grade V3c. The class B strapping need not apply. Contents shall be cushioned, blocked, braced, and anchored to prevent movement.

5.2.3 Commercial. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition. Containers and packing shall comply with carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The trap covered by this specification can be used in dual-purpose applications. By automatic operation, the trap can be used either to drain water or light oils from pressurized air or gas systems, or to vent air and other noncondensable gases from systems that use water, gasoline, or light oils under pressure.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article sample is required for inspection and approval (see 3.2, 4.3.1, and 6.3).
(c) Pipe size of 18NPT female thread required (see 3.7.7).
(d) If different water discharge capacity is required (see 3.8.3.1).
(e) If different air venting capacity is required (see 3.8.3.2).
(f) Treatment and painting, if different (see 3.9).
(g) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Defense Acquisition Regulations. The first article should be a first production item consisting of one complete float trap unit or it may be a standard production item from the contractor's current inventory as specified in 4.3.1. The contracting officer should include specific instructions in all acquisition instruments regarding arrangement for examinations, tests, and approval of the first article.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

☆U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/2212

Preparing activity:

Navy - YD

Project No. 4730-N515

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:46Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. 4730-N515

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-T-21092C(YD) TRAPS, AIR AND WATER, BALL FLOAT

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW:

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

S/N 0102-LF-001-4260

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:46Z
Check the source to verify that this is the current version before use.